

# CS 253: Data Structures and Algorithms

## Randomized Final Study Problem

### Dynamic Programming

#### 0/1 Knapsack

Given a knapsack of capacity 10 with weights  $w = \{4, 6, 1, 5, 2\}$  and values  $v = \{14, 9, 9, 10, 5\}$ , complete the 0/1 knapsack dynamic-programming table and determine the optimal value.

| $i \setminus c$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|-----------------|---|---|---|---|---|---|---|---|---|---|----|
| 0               | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  |
| 1               | 0 |   |   |   |   |   |   |   |   |   |    |
| 2               | 0 |   |   |   |   |   |   |   |   |   |    |
| 3               | 0 |   |   |   |   |   |   |   |   |   |    |
| 4               | 0 |   |   |   |   |   |   |   |   |   |    |
| 5               | 0 |   |   |   |   |   |   |   |   |   |    |

#### Instructor Reference

Greedy-by-ratio value: 28. Optimal DP value: 33.

| $i \setminus c$ | 0 | 1 | 2 | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|-----------------|---|---|---|----|----|----|----|----|----|----|----|
| 0               | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 1               | 0 | 0 | 0 | 0  | 14 | 14 | 14 | 14 | 14 | 14 | 14 |
| 2               | 0 | 0 | 0 | 0  | 14 | 14 | 14 | 14 | 14 | 14 | 23 |
| 3               | 0 | 9 | 9 | 9  | 14 | 23 | 23 | 23 | 23 | 23 | 23 |
| 4               | 0 | 9 | 9 | 9  | 14 | 23 | 23 | 23 | 23 | 24 | 33 |
| 5               | 0 | 9 | 9 | 14 | 14 | 23 | 23 | 28 | 28 | 28 | 33 |